

# Jai Sinha

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## EDUCATION

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### University of Wisconsin–Madison

Aug 2021 – May 2025

*Bachelor of Science in Computer Science*

*Madison, WI*

*Related Courses: Linear Algebra, Applied Statistics, User Interfaces, Computer Engineering, Computer Graphics, Machine Organization, Algorithms, Artificial Intelligence, Big Data, Database Implementations*

## EXPERIENCE

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### Software Engineering Intern

May 2024 – Aug 2024

*Tesla*

*Palo Alto, CA*

- Used Go routines to remodel tester data upload process, reducing cycle time by over 10% and preventing massive data losses from occurring on manufacturing lines.
- Led Go software development for major new tester, communicating with Battery and Firmware teams to ensure full coverage of a detailed task plan, delivering 11 test states well ahead of deadlines.
- Created tool to individually validate all 14 functionalities of various off-the-shelf hardware tools used in tester, using Go channels to pass messages through a web socket between the Go backend and React frontend.
- Worked closely with Firmware team to bring 20+ new functionalities to widely shared toolset abstracting CAN messages and parsing bit-level payloads.

### Software Engineering Intern

May 2023 – Aug 2023

*Tesla*

*Palo Alto, CA*

- Developed LabVIEW software for high-volume tester, shortening tester duplication process by 30+ minutes, and supported debugging and error analysis across a 15-hour time zone difference.
- Automated data acquisition setup by creating LabVIEW-based driver using NI DAQmx libraries, and added support for XML configuration files to further simplify the setup process.
- Supported 50+ hours of tester pre-ship software validation and hardware troubleshooting, learning to solder, read wiring diagrams, use digital multimeters, and coordinate with other teams.
- Created template dashboard using internal SQL-based tools to visualize tester performance based on metrics such as overall yield, Pareto failure analysis, and more, with process duplication possible across 1000s of testers.

## PROJECTS

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### Car Scraper | *Python, Playwright, Quart, React, PostgreSQL, Docker*

June 2025 – Aug 2025

- Built a full-stack web app to aggregate and filter live car auction listings from major platforms.
- Implemented a responsive React frontend with advanced keyword-based search and filtering.
- Developed an asynchronous RESTful API in Quart with secure auth and user-specific features like saved listings, all persisted in PostgreSQL
- Engineered an async Playwright scraper to fetch listing data concurrently, reducing scrape time by over 70%.
- Designed high-performance SQL queries to filter 1,000+ listings with sub-second response times.
- Containerized the entire system with Docker and Docker Compose for streamlined development, testing, and scalable deployment.

### Portfolio Site | *React, HTML/CSS, GitHub Pages*

May 2025

- Built and deployed a photography portfolio using React, featuring a dynamic homepage and photo gallery.
- Implemented a responsive masonry-style grid with lazy-loaded thumbnails linked to high-resolution versions, optimizing load time without sacrificing image quality.
- Deployed to GitHub Pages with minimalist design optimized for fast load times.

## TECHNICAL SKILLS

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**Languages:** Go, Python, Javascript, SQL (Postgres), C/C++, Java, Rust

**Frameworks:** React, Quart, Flask, Bootstrap, Playwright, Apache Spark

**Developer Tools:** Git, Docker, Google Cloud Platform, VS Code, Vite, JupyterLab